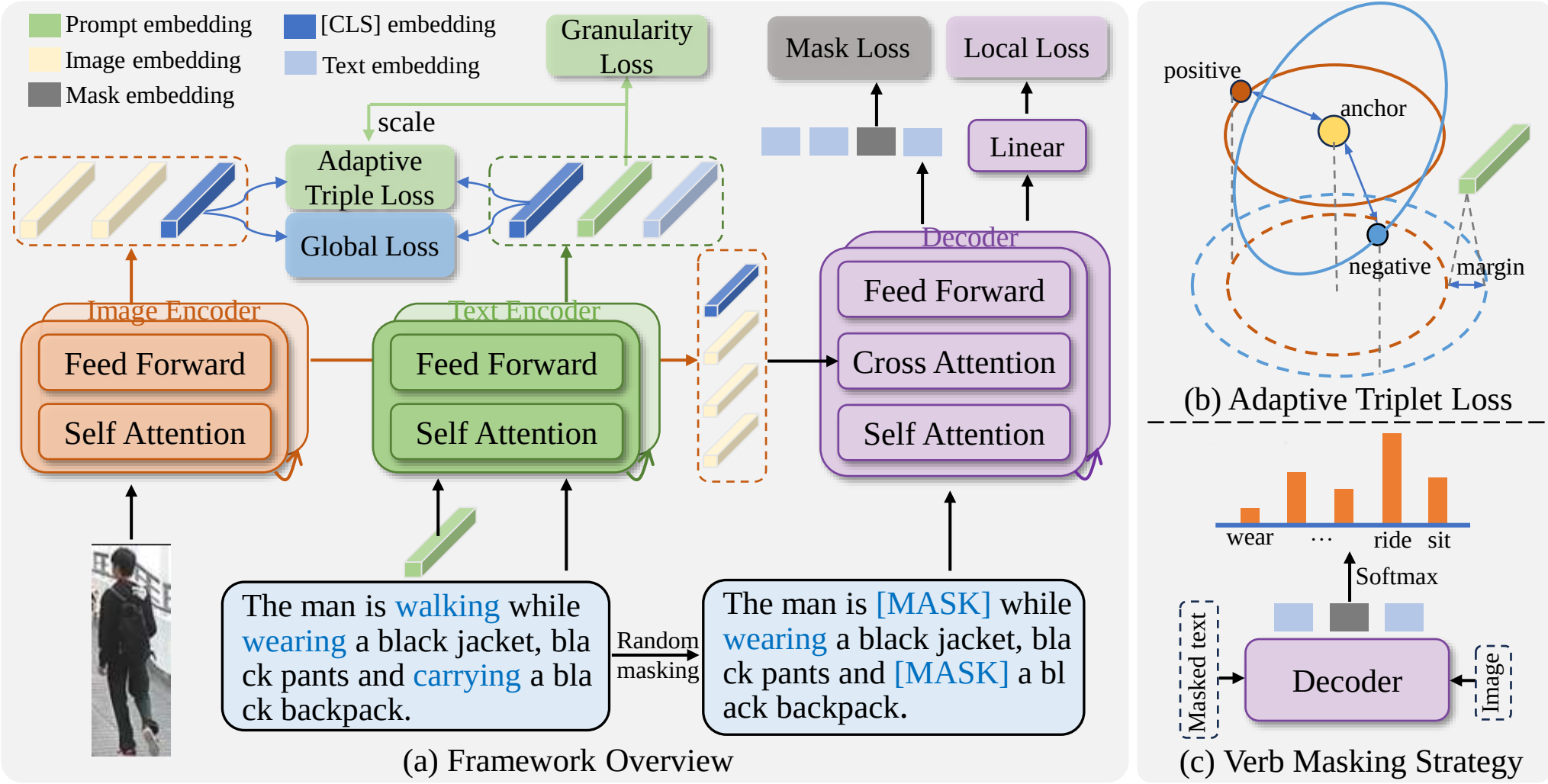

Prompt embedding
Image embedding
Mask embedding
[CLS] embedding
Text embedding
Granularity Loss
Mask Loss
Local Loss
scale
Adaptive Triple Loss
Global Loss
Linear
Decoder
Image Encoder
Feed Forward
Self Attention
Text Encoder
Feed Forward
Self Attention
Feed Forward
Cross Attention
Self Attention
The man is walking while wearing a black jacket, bla ck pants and carrying a bla ck backpack.
Random masking
The man is [MASK] while wearing a black jacket, bla ck pants and [MASK] a bl ack backpack.
(a) Framework Overview
positive
anchor
negative
margin
(b) Adaptive Triplet Loss
wear
ride
sit
Softmax
Masked text
Decoder
Image
(c) Verb Masking Strategy